

Phase 2: Requirement Analysis

Requirements Specification Document

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Software Dependencies

- **Operating System:** Windows 10/11, macOS, or Ubuntu Linux.
- **Runtime Environment:** Python 3.12 or higher.
- **External Packages:** Flask, pandas, numpy, scikit-learn, xgboost, seaborn, matplotlib (managed via `requirements.txt`).

Hardware Configurations

- **Processor:** Intel i5 / AMD Ryzen 5 or higher (Multi-core preferred for faster hyperparameter tuning).
- **RAM:** Minimum 8 GB (16 GB recommended during the RandomizedSearchCV phase).
- **Storage:** Minimum 2 GB of free disk space for datasets, virtual environments, and dependencies.

Data Specifications

The system utilizes the `loan_prediction.csv` dataset.

- **Total Records:** 614 entries.
- **Total Columns:** 13 features.

- **Datatypes:** 12 features are strictly encoded as `int64` (e.g., Gender, Married, Education, ApplicantIncome). 1 feature is `str` (Loan_ID, which is dropped prior to model training).
- **Target Variable:** `Loan_Status` (Integer encoded: 1 for Approved, 0 for Rejected).